

1.15 Solution: Take the functional given by Eq. (1.67) and generalize to multiple surfaces scenario, we have

$$I[\psi, \{f\}] = \frac{1}{2} \int_V \nabla \psi \cdot \nabla \psi d^3x - \oint_V g \psi d^3x - \sum_{i=1}^N \int_{S_i} f_i \psi da_i,$$

subject to the constraints

$$\oint_{S_i} f_i da_i = Q_i, \quad \text{for } i = 1, \dots, N.$$

To find the minimum of the functional under the constraints, we can introduce Lagrange multipliers λ_i for every surface, and the corresponding Lagrangian is

$$\mathcal{L}[\psi, \{f\}, \{\lambda\}] = \frac{1}{2} \int_V \nabla \psi \cdot \nabla \psi d^3x - \oint_V g \psi d^3x - \sum_{i=1}^N \int_{S_i} f_i \psi da_i + \sum_{i=1}^N \lambda_i \left(\int_{S_i} f_i da_i - Q_i \right).$$

The variation of the Lagrangian is given by Eq. (1.68), with extra terms,

$$\begin{aligned} \delta \mathcal{L}[\psi, \{f\}, \{\lambda\}] &= \int_V [-\nabla^2 \psi - g] \delta \psi d^3x + \sum_{i=1}^N \oint_{S_i} \left(\frac{\partial \psi}{\partial n} - f_i \right) \delta \psi da_i \\ &\quad - \sum_{i=1}^N \oint_{S_i} \delta f_i (\psi - \lambda_i) da_i + \sum_{i=1}^N \delta \lambda_i \left(\int_{S_i} f_i da_i - Q_i \right). \end{aligned}$$

For $\delta \mathcal{L}$ to vanish, we need to have

$$\nabla^2 \psi = -g \text{ within } V, \quad \frac{\partial \psi}{\partial n} = f_i \quad \text{and} \quad \psi = \lambda_i \text{ on } S_i.$$

Therefore, we have established that every surface is equipotential.